Categories, Proofs, and Processes:
Lecture 17 Adjunctions

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8 Monoidal Categories

Definition 8.1 (Informal)**.** Monoidal categories are categories with a "parallel" composition $\otimes$.

8.1 Motivating Examples

Example 8.2 $(\mathbf{Set}, \times)$**.**

$$f : A \to B, \quad g : C \to D$$

$$f \times g : A \times C \to B \times D$$

$$(f \times g)(a, c) = (f(a), g(c))$$

$$\begin{tikzcd}
A \arrow[d, "f"'] & A \times C \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] \arrow[d, dashed, "f \times g"] & C \arrow[d, "g"] \\
B & B \times D \arrow[l, "\pi_1'"] \arrow[r, "\pi_2'"'] & D
\end{tikzcd}$$

Any $\mathcal{C}$ with products

$$f \times g := \langle f \circ \pi_1, g \circ \pi_2 \rangle$$

$$\begin{aligned}
(f' \times g') \circ (f \times g) &= (f' \times g') \circ \langle f \circ \pi_1, g \ circ \pi_2 \rangle \\
&= \langle f' \circ f \circ \pi_1, g' \circ g \circ \pi_2 \rangle \\
&= (f' \circ f) \times (g' \circ g) \quad \text{(interchange law)}
\end{aligned} \tag{1}$$

Example 8.3 $(\mathbf{Set}, +)$**.** $+$ is the disjoint union.

$$f : A \to B, \quad g : C \to D$$

$$f + g : A + C \to B + D$$

$$(f + g)(a, c) = \begin{cases} f(x) & \text{if } x \in A \\ g(x) & \text{if } x \in C \end{cases}$$

In $\mathcal{C}$ with coproducts:

$$f + g = [\iota_1 \circ f, \iota_2 \circ g]$$

"Co-tupling of maps"

$$\begin{tikzcd} A \arrow[r, "\iota_1"] \arrow[d, "f"'] & A + C \arrow[d, dashed, "f+g"] & C \arrow[l, "\iota_2"'] \arrow[d, "g"] \\ B \arrow[r, "\iota_1'"'] & B + D & D \arrow[l, "\iota_2'"] \end{tikzcd}$$

$$(f' + g') \circ (f + g) = (f' \circ f) + (g' \circ g)$$

8.2 Non-Examples

Example 8.4 $(\mathbf{Rel}, \times)$**.** $\times$ is not a product or a coproduct since relations are non-separable. Let

$$\mathbb{B} = \{0, 1\}$$

$$S = \{* \mapsto (0,0), * \mapsto (1,1)\}$$

$$R = \{* \mapsto 0, * \mapsto 1\}$$

$$\begin{tikzcd} \mathbb{B} & \mathbb{B} \times \mathbb{B} \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & \mathbb{B} \\ & \{*\} \arrow[ul] \arrow[u, "S"'] \arrow[ur] & \end{tikzcd}$$

$$S' = \begin{cases} * \mapsto (0,0) \\ * \mapsto (0,1) \\ * \mapsto (1,0) \\ * \mapsto (1,1) \end{cases}$$

$S' \neq S$, no unique relation.

Example 8.5 $(\mathbf{Vect}_{\mathbb{C}}, \otimes)$**.** $\otimes$ is the tensor product.

8.3 Definitions

Definition 8.6. A monoidal category $(\mathcal{C}, \otimes, I)$ is a category $\mathcal{C}$ with a functor $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ called the monoidal product, an object $I \in \mathrm{Ob}(\mathcal{C})$ called the unit and natural isomorphisms

$$\alpha_{A,B,C} : (A \otimes B) \otimes C \xrightarrow{\sim} A \otimes (B \otimes C)$$

$$\lambda_A : I \otimes A \xrightarrow{\sim}, \quad \rho_A : A \otimes I \xrightarrow{\sim} A$$

satisfying 2 coherence conditions.

Example 8.7.

$$\{(a,b), c | a, b, c\} \cong \{a, (b,c) | a, b, c\}$$

Definition 8.8. A monoidal category is strict if α, λ, ρ are all identities.

Claim 8.9. *Any monoidal category is equivalent to a strict category.*

8.4 Aside: Term rewrite systems

Definition 8.10. A signature

$$\Sigma = \{\otimes : 2, I : 0\}$$

where $\otimes$ is a symbol, $2, 0$ are arities, with arity $0 \Rightarrow$ constant.

Definition 8.11. A term $t \in T_\Sigma(V)$ for some set V (variables) is a tree whose nodes are in Σ and leaves are in V.

Example 8.12.

$$V = \{x, y, z\}, \quad T_\Sigma V \ni \begin{cases} (x \otimes x) \otimes x \\ I \otimes I \\ (x \otimes y) \otimes z \\ \ldots \end{cases}$$

Definition 8.13. A rewrite rule is a pair of terms

$$(x \otimes y) \otimes z \xrightarrow{R_1} x \otimes (y \otimes z)$$

$$I \otimes x \xrightarrow{R_2} x$$

$$x \otimes I \xrightarrow{R_3} x$$

Rewrite rules can be applied by

1. substituting variables
2. replacing a sub-term

For

$$x \mapsto a, y \mapsto b, z \mapsto I$$

$$(a \otimes b) \otimes I \xrightarrow{R_1'} a \otimes (b \otimes I)$$

$$((a \otimes b) \otimes I) \otimes d \xrightarrow{R_1' \otimes d} (a \otimes (b \otimes I)) \otimes d$$

Definition 8.14. "Good" rewrite systems are:

- terminating (all sequences $t \to \ldots \to t'$ are finite)
- confluent:

$$\forall t_1, t_2 . t \rightrightarrows\rightrightarrows\rightrightarrows t_1, t_2 \exists t_3 . t_1, t_2 \rightrightarrows\rightrightarrows\rightrightarrows t_3$$

Terminating + confluent $\implies$ unique normal forms.

Example 8.15 (Confluence)**.**

$$\begin{tikzcd} (x \otimes I) \otimes y \arrow[r] \arrow[d] & x \otimes (I \otimes y) \arrow[dl] \\ x \otimes y & \end{tikzcd}$$

Definition 8.16. "Coherence = confluence + commuting diagrams"

$$
\begin{tikzcd}
 & & T_1 \arrow[dr] & & \\
 & {} \arrow[ur] & & {} \arrow[dr] & \\
T \arrow[ur] \arrow[dr] & & & & T_3 \\
 & {} \arrow[dr] & & {} \arrow[ur] & \\
 & & T_2 \arrow[ur] & &
\end{tikzcd}
$$

commutes.

Definition 8.17 (MacLane's coherence conditions)**.**

$$
\begin{tikzcd}
(A \otimes I) \otimes B \arrow[d, "\alpha"'] \arrow[dr, "\rho \otimes B"] & \\
A \otimes (I \otimes B) \arrow[r, "A \otimes \lambda"'] & A \otimes B
\end{tikzcd}
$$

$$
\begin{tikzcd}
 & (A \otimes B) \otimes (C \otimes D) \arrow[rr, "\alpha_{A \otimes B, C, D}"] & & A \otimes (B \otimes (C \otimes D)) \\
((A \otimes B) \otimes C) \otimes D \arrow[ur, "\alpha"] \arrow[dr, "\alpha \otimes D"'] & & & \\
 & (A \otimes (B \otimes C)) \otimes D \arrow[r, "\alpha"'] & A \otimes ((B \otimes C) \otimes D) \arrow[uur, "A \otimes \alpha"'] &
\end{tikzcd}
$$

(Δ) (top) and ($\pentagon$) (bottom) coherence.

Theorem 8.18 (Coherence)**.** *All formal* diagrams of λ, ρ, α commute.*